

Solution Requirements

Date	28 June 2026
Team ID	SWTID-2026-2260
Project Name	Smart Lender
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Bank credit officers and financial analysts securely log in using username and password to access the Smart Lender system.	High
2	Authorization levels	The system provides role-based access. Credit Officers can evaluate loan applications, Financial Analysts can view reports, and Administrators can manage users and ML models.	High

3	External interfaces	The application interacts with the trained XGBoost model, Flask backend, IBM Cloud platform, and applicant database for prediction and storage.	High
4	Transactions processing	The system accepts applicant details, validates the data, predicts loan approval using the ML model, and stores the prediction results in the database.	High
5	Reporting	<i>Generate loan approval reports, prediction summaries, applicant history, approval/rejection statistics, and model performance reports.</i>	Medium
6	Business rules, etc.	<i>Applicants with good credit history and sufficient income have higher approval probability. Loan eligibility is determined based on applicant details and the trained XGBoost model.</i>	High
7	Compliance to laws or regulations	<i>Ensure applicant data privacy and secure storage by following banking security guidelines and applicable data protection standards.</i>	High
8	Other	<i>Provide a user-friendly web interface that instantly displays loan approval or rejection after prediction.</i> <i>Medium</i>	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric/ Acceptance Criteria
1	Performance & Speed	<i>The system should generate loan predictions quickly after applicant details are submitted.</i>	<i>Prediction completed within 2 seconds</i>
2	Scalability	<i>The application should support multiple users submitting loan applications simultaneously.</i>	<i>Supports 1000+ concurrent users</i>
3	Security & Data Privacy	<i>Applicant information must be protected using secure authentication, encrypted communication, and secure database storage.</i>	HTTPS, encrypted data transmission, secure login
4	Reliability & Availability	<i>The Smart Lender application should remain available for banking operations with minimal downtime.</i>	99.9% uptime
5	Usability & Accessibility	<i>The application should have a simple, responsive interface that is easy for applicants and banking staff to use.</i>	Responsive design and intuitive navigation

6	Other	<i>The system should be easy to maintain, update with new ML models, and deploy on IBM Cloud or other cloud platforms.</i>	Modular Python-Flask architecture
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